Note: This document is maintained by PawCore Manufacturing Quality Control Department.

Last Review Date: September 1, 2024

Next Review Due: March 1, 2025v

PAWCORE MANUFACTURING QUALITY CONTROL STANDARDS

Version 2.4 (Updated September 2024)

1. PRODUCT QUALITY CONTROL OVERVIEW
1.1 SmartCollar Device Classification

Product Category: Pet Wearable Electronics
Safety Classification: Class II
Water Resistance Rating: IP67
Operating Temperature Range: -10°C to 45°C
Battery Type: Lithium-Ion 3.7V

1.2 General Testing Requirements

Visual Inspection
Functional Testing
Safety Testing
Durability Testing
Battery Performance Testing
Water Resistance Testing

2. MANUFACTURING QUALITY CONTROL PROCEDURES
2.1 Component Testing
Battery Assembly

Voltage Output Test (3.6V - 3.8V)
Charge Cycle Test (minimum 500 cycles)
Short Circuit Protection Test
Temperature Monitor Test (0°C - 45°C)
Basic Load Test (24-hour duration)

Circuit Board Assembly

Power Draw Test
Signal Strength Test
Component Placement Verification
Solder Joint Inspection
Basic Functionality Test

2.2 Assembly Quality Control

Housing Seal Inspection
Screw Torque Verification
Button Function Test
LED Indicator Test
Basic Water Resistance Check (30-minute submersion)

3. ENVIRONMENTAL TESTING STANDARDS
3.1 Temperature Testing

Cold Test (-10°C for 4 hours)
Heat Test (45°C for 4 hours)
Temperature Cycle Test (5 cycles)

3.2 Water Resistance Testing

Light Rain Simulation (IPX4)
Water Jet Test (IPX5)
Temporary Immersion (IPX7)
Basic Splash Test

3.3 Mechanical Testing

Drop Test (1.5m height)
Vibration Test
Bend Test
Clasp Durability (5000 cycles)

4. QUALITY CONTROL CHECKPOINTS
4.1 Pre-Assembly QC

Component Verification
Material Inspection
Batch Tracking Setup
Tool Calibration Check

4.2 In-Process QC

Assembly Line Inspection Points
Functional Testing Stations
Basic Environmental Checks
Visual Inspection Points

4.3 Final QC

Full Functionality Test
Basic Environmental Test
Packaging Inspection
Documentation Review

5. TEST RESULT DOCUMENTATION
5.1 Required Documentation

Batch Number
Test Date
Operator ID
Test Station ID
Pass/Fail Status
Basic Environmental Conditions

Room Temperature
Humidity (if available)

5.2 Test Result Parameters
Battery Performance

Initial Voltage: 3.7V ±0.1V
Charge Time: < 2 hours
Discharge Test: > 72 hours
Cycle Test: Pass/Fail

Water Resistance

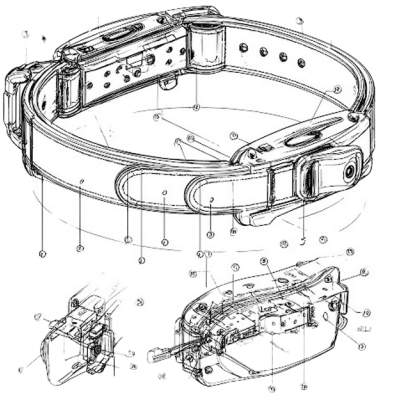
30-minute submersion test
Basic spray test
Seal inspection

6. QUALITY CONTROL SIGN-OFF
6.1 Required Approvals

Assembly Line Supervisor
Quality Control Inspector
Production Manager

6.2 Documentation Storage

Test Results: 3 years
Batch Records: 5 years
Failure Reports: 3 years



7. REVISION HISTORY
Version 2.4 (September 2024)

Updated battery test parameters
Added basic water resistance check
Updated documentation retention policy

Version 2.3 (June 2024)

Added temperature cycle test
Updated assembly line inspection points

Version 2.2 (March 2024)

Added batch tracking requirements
Updated safety classification

PawCore
PET HEALTH & FITNESS TRACKER